Generative Molecular Design

Proven with NVIDIA GenMol — on a single GPU

We fine-tuned NVIDIA's GenMol model on a real dataset and it generated valid, novel, drug-like molecules aligned to the target chemistry.

The approach works — and it's ready to apply to your data.

THE PROBLEM

The Challenge in Early Discovery

- **Finding new molecules is slow, costly, and largely trial-and-error.**
- Chemists can only explore a tiny fraction of possible chemical space by hand.
- Each dead-end candidate consumes time, lab resources, and budget.
- Teams need novel, viable starting points — faster and at lower cost.

Why it matters

Early-stage ideation is a major bottleneck in the discovery timeline.

Compressing it means more shots on goal, sooner — with the same team.

The Opportunity: Generative AI for Chemistry

- **Generative models can propose brand-new, viable molecules on demand.**
- Explore vastly more of chemical space than manual design allows.
- Steer generation toward the chemistry you care about (your target domain).
- Turn early ideation from a slow bottleneck into an on-demand capability.

NVIDIA GenMol is a pretrained generative model for molecules — we adapt it to a specific domain and put it to work.

THE CAPABILITY

More Than One Trick — What GenMol Can Do

De novo design

Invent brand-new molecules from scratch

Scaffold decoration

Grow R-groups on a fixed core structure

Linker design

Connect two fragments into one molecule

Motif extension

Build outward from a known active motif

Lead optimization

Improve a molecule while keeping it similar

Goal-directed

Steer toward property targets (potency, QED...)

One model spans the whole hit-to-lead workflow — published validity ~100% and novelty ~90% on de novo generation.

Why Not Just Use It Off-the-Shelf?

Off-the-shelf model

- A generalist trained on broad, generic chemistry
- Generates good molecules — but not your molecules
- No knowledge of your target domain or objectives

Fine-tuned on your data

- Keeps the base model's broad chemical knowledge
- Adds focus on your target chemistry — measurably
- Our PoC moved output +43% closer to the target domain
- Adapts in hours on 1 GPU — no training from scratch (8 GPUs)

Our Approach — A Repeatable Method

1 · Prepare Data

Clean and structure a target molecular dataset

2 · Fine-tune Model

Adapt NVIDIA GenMol to the target chemistry

3 · Generate & Evaluate

Produce new molecules and measure quality objectively

Disciplined and reproducible: every run is seeded, tracked, and automatable — the same method applies to any dataset or objective.

THE DATA

What We Trained On

Base model (by NVIDIA)

- Pretrained on a large public drug-like molecule corpus
- Encodes broad, general chemical knowledge
- Trained once by NVIDIA on 8 GPUs — we reuse it

Our fine-tuning data

- Delaney / ESOL — 1,056 curated drug-like molecules
- Split into 951 train / 105 held-out for honest evaluation
- Used only the molecular structures — our target domain
- A public stand-in for the customer's proprietary dataset

A broad generalist, focused on a specific chemistry — on a single GPU.

SETTING THE BAR

What Success Looks Like

Valid	Novel	Drug-like	On-domain
Chemically real, usable molecules	Genuinely new — not copies	Viable candidate properties	Aligned to the target chemistry

We agreed these criteria up front and measured every one — so results are judged objectively, not by eye.

PROOF POINT

The Results — It Works

97.5%

of generated molecules are chemically valid

85%

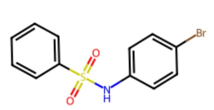
are genuinely novel — new molecules

+43%

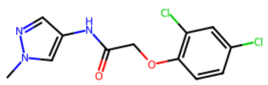
closer to the target chemistry (similarity)

- More of the output is drug-like (drug-like fraction rose from 34% to 42%).
- The model measurably learned the target chemistry — its output moved toward the desired molecules.
- **All achieved while keeping quality high — a successful, credible adaptation.**

Molecules the Model Invented



QED 0.95 | SA 1.48 | sim 0.28



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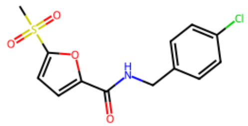
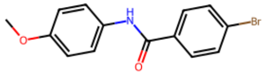
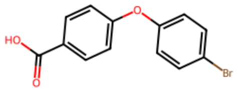


QED 0.94 | SA 2.07 | sim 0.25



QED 0.94 | SA 1.34 | sim 0.25



QED 0.93 | SA 1.47 | sim 0.28

- **These are new, valid, drug-like structures the model created.**
- None are copied from the training data.
- Each scores well on drug-likeness and ease of synthesis.
- This is the tangible output a discovery team would triage.

Efficient and Ready to Scale

1 GPU	Slurm	Repeatable
ran on a single NVIDIA A100	standard cluster scheduling	seeded, tracked, automatable

- Modest, standard infrastructure — cost-efficient to run and easy to budget.
- Scales cleanly to larger datasets and automated experiment sweeps.
- **Every run is tracked and reproducible — no black-box, no vendor lock-in surprises.**

WHY YOU CAN TRUST IT

A Credible Study, Not a Demo

- **We measured real trade-offs, not just the wins.**
- Specializing the model narrows its focus — a normal, tunable effect we quantified.
- We validated on molecules the model never saw during training (held-out set).
- Results are honest, reproducible, and backed by objective metrics.

This rigor is what separates a proof of concept from a demo — and what de-risks scaling it.

STRAIGHT TALK

Known Limits — and How We Address Them

Limitation	How we address it
Broad, public dataset used	Your proprietary, focused data should give a sharper, more useful fit
Single evaluation run	A pilot confirms results across multiple runs + an independent test set
Proxy chemistry measures	Add your real endpoints: potency, ADMET, synthesizability
Broad chemical domain	Target a specific series or objective for stronger, actionable results

YOUR OPPORTUNITY

What This Means for You

- **The same pipeline runs on your proprietary data and your objectives.**
- Target your chemistry — potency, solubility, ADMET, or a specific series.
- Generate novel candidate molecules on demand, evaluated against your criteria.
- Built on NVIDIA infrastructure you can scale as needs grow.

This proof of concept becomes your capability — applied to the molecules that matter to your business.

The Ask: A Scoped Pilot

What we need from you

- A representative dataset for your target chemistry
- Agreed success metrics
- An executive sponsor

What you get

- Domain-specific generated candidates
- Objective evaluation against your criteria
- A scalable capability on NVIDIA infrastructure

Next step: agree the dataset, metrics, and timeline — and start the pilot.